MILITARY SPECIFICATION

MOUNTER-DEMOUNTER, PNEUMATIC TIRE, STATIONARY

This specification is approved for use by all Deparment and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of stationary, power-driven, mounter-demounter that changes pneumatic tires from wheeled vehicles.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-621	-	Box, Wood, Nailed and Lock Corner
PPP-B-636	-	Box, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-T-60	-	Tape, Packaging, Waterproof

MILITARY

MIL-P-116	-	Preservation, Methods of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-C-5501	-	Cap and Plug, Protective, Dust and Moisture Seal
MIL-M-7298	-	Manuals, Technical: Commercial Equipment
MIL-L-8362	-	Lubricant, Tire and Rim, Demounting

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:19Z
Check the source to verify that this is the current version before use.

MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts; Packaging of
MIL-P-19834 - Plate, Identification, Metal Foil, Adhesive Backed
MIL-C-52950 - Crate, Wood, Open and Covered

STANDARDS

FEDERAL

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government
FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With appropriate Test Methods
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking

DOD-STD-2000-4 - General Purpose Soldering Requirements for Electrical and Electronic Equipment

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

201.02 - Tooth Proportions for Coarse-Pitch Involute Spur Gears
390.03 - Gear Classification Manual
600.01 - Metric Gears

(Application for copies should be addressed to American Gear Manufacturer's Association, One Thomas Circle, Washington, DC 20005.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C1 - National Electrical Code
ANSI/ASTM E 380 - Metric Practice

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:19Z
Check the source to verify that this is the current version before use.

Electrical Standards
MG-1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 East 44th Street, New York, NY 10017.)

INTERNATIONAL STANDARDS ORGANIZATION (ISO)

ISO Standard 1000 - SI Units and Recommendation for the Use of their Multiples and of Certain Other Units Second Edition Erratum 1981

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, one complete mounter-demounter shall be subjected to first article testing and inspection. (see 4.3, 6.2 and 6.4.)

3.2 Design. The mounter-demounter may be of any design that meets the requirements as stated herein.

3.2.1 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the tire support industry. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.2.2 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.2.3 Measurement commonality. The US Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the mounter-demounter, its components, and all ancillary equipment. Whichever system of measurement the manufacturer elects to incorporated, US or SI, that system shall be incorporated throughout the moutner-demounter, its components, and all ancillary equipment. In this specification, all measurements, dimensions, sizes and

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:19Z
Check the source to verify that this is the current version before use.

capacities are given in the US Customary System of Units. The measurements may be converted to the SI Units through the use of conversion factors and methods specified in FED-STD-376.

3.3 Construction. The mounter-demounter shall be complete so that, when installed or connected to its specified source of power (see 3.2.1), it can be used for any operation for which designed. Construction shall be free from any characteristic or defect that would prevent the mounter-demounter from performing any of the requirements specified herein.

3.3.1 Castings and forgings. All castings shall be free of blow holes, porosity, hard spots, cracks, and other surface defects. Forgings shall be free from scale, cold shuts, mismatching, and other surface defects. Strength and other essential properties of the castings and forgings shall meet the performance requirements specified herein.

3.3.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations, however, shall not be employed as repair measures for defective parts.

3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be swagged, peened, staked, or otherwise deformed to the extent that the parts cannot be readily disassembled and reassembled.

3.3.4 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.3.5 Gears. All gears and pinions used in the mounter-demounter or its components shall be fabricated in either the English (USA) system or the Metric (SI) system of measurement. English gears shall conform to AGMA 390.03. Spur gears shall conform to AGMA 201.02. Metric gears shall comply with the SI Units, International Standards Organization (ISO) Standard 1000, American National Standard ANSI/ASTM E 380, and American Gear Manufacturers Association 600.01.

3.3.6 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible. Lubrication reservoirs, if applicable, shall have means for checking levels.

3.3.7 Safety devices. Each mounter-demounter shall be furnished with safety devices. Parts which are hazardous to the operator shall be guarded. Protection against electrical shock shall be provided. When specified (see 6.2), special safety devices shall be provided as required by the procuring activity. Sharp corners and edges that present hazards shall be removed or shielded.

3.3.8 Interchangeability. All replaceable parts shall be designed and manufactured to such standards as shall permit replacement and adjustment without the need for modification of the parts or the mounter-demounter.

3.3.9 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:19Z
Check the source to verify that this is the current version before use.

3.3.10 Painting. unless otherwise specified (see 6.2), painting and finishing shall be in accordance with standard commercial practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall be clean and free of all foreign matter detrimental to painting and at least one coat of primer and one coat of enamel, or equivalent applied.

3.4 Performance. The mounter-demounter shall successfully perform all operations specified herein with no more than one operator required to accomplish the tasks, in the time frame specified, with no more equipment than is to be supplied with the mounter-demounter, and with no visible damage to the mounter-demounter or the wheels and tires. The tire mounter-demounter shall successfully meet all performance requirements stated herein under its own power. It shall not require assistance from the machine operator or other personnel in the form of leverage bars or other contrivances to aid or cause the mounter-demounter to rotate and thereby accomplish lifting the bead foot completely over and off of the rim of the wheel. (This requirement does not apply to the tire irons used to lift the bead for inserting powered bead lifting tools or to other tools such as pliers that are used to hold the bead after it has been lifted.)

3.4.1 Operating time. The mounter-demounter shall completely demount and remount each tire specified herein in no more than 20 minutes for the most difficult tire, and less time for less difficult tires. The 20 minute time frame begins with a deflated tire on a wheel with the bead not yet broken, setting on the floor next to the mounter-demounter, but not touching it. The time frame ends with the reassembled wheel and tire back on the floor, not touching the mounter-demounter, the bead not yet set, and the tire ready for reinflation. (This time frame does not include changing valve stems.)

3.4.2 Tire and wheel damage. The mounter-demounter shall successfully meet all performance requirements specified herein and shall not damage any and all wheels or tires specified herein as a result of the bead breaking and mounting-demounting operations. There shall be no visible evidence of damaged rubber, sidewall, bead foot, or damaged wheel parts including dents to any part of the wheel or scratches that will cause loss of air pressure. There shall be no permanent visible deformation of metal in the wheels beyond its initial manufactured state.

3.4.3 Tire and rim lubricant. Tire and rim lubricant in accordance with MIL-L-8362 shall be provided and delivered in two gallon quantities with each mounter-demounter. An applicator shall be provided and a devoted resting place for the applicator and at least one pint of lubricant shall be provided on the machine in an easily reached position. At least one refillable bottle no smaller than one pint in holding capacity shall be provided to hold the lubricant for use and shall fit the devoted storage location on the machine.

3.4.4 Tire capacity. The mounter-demounter shall be capable of mounting and demounting both tube and tubeless type pneumatic tires 7.00-15 through 14.00-24 and 18-22.5, 2 ply through 22 ply and all other tires specified herein.

3.4.4.1 M198. The mounter-demounter shall successfully break the bead and demount a used Goodyear Supersingle KK9592B 16.5-19.5 16 Ply rated tubeless tire from the wheel used on the M198 Howitzer. The M198 Howitzer may be located at many Active Army, Reserve Army or National Guard installations. After breaking the bead and demounting the used tire, the mounter-demounter shall successfully mount a new Goodyear Supersingle KK9592B on the same rim.

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:19Z
Check the source to verify that this is the current version before use.

3.4.4.2 M747. The mounter-demounter shall successfully break the bead and demount a used Firestone Tire Part Number 2F-29G-22-C-95 from either of two types of special military wheels found on the M747 Heavy Equipment Trailer. The tire is a 15-19.5, 14 ply rated. The wheel and tire assembly may be located at most state National Guard maintenance shops and at many US Army Maintenance Depots. After breaking the bead and dismounting the old tire, the mounter-demounter shall successfully mount a new Firestone 2F-29G-22-C-95 on the same rim.

NOTE: Due to uncontrollable circumstances, some M747 trailers may have non-standard or unauthorized tires mounted to their wheels. Check the tires for correct identification before completing any transactions to borrow them from military installations.

3.4.4.3 M915A1. The mounter-demounter shall break the bead and demount a used Goodyear Tire Number 135-802-993 from a Firestone Rim Number 27404. The tire is an 11R-22.5, 14 ply rated and the rim is an 11.00 X 22.5. The wheel and tire assembly is normally found on the M915A1 truck tractor and can be located at most state National Guard Headquarters. After successfully breaking the bead and demounting this tire from this rim the mounter-demounter shall successfully mount a new Goodyear Number 135-802-993 on the same rim.

3.4.4.4 CUCV-I. The mounter-demounter shall successfully break the bead and demount a used 10R15C on/off road tire from a General Motors Wheel Number 1406337. For information purposed, the wheel is a 15 X 8.0 with six hole lug pattern having a 5.5 inch diameter bolt circle and a .66 inch offset. The wheel and tire assembly may be obtained from any commercial source and may also be found at most active Army installations and some National Guard units. After successfully breaking the bead and demounting this tire the mounter-demounter shall successfully mount a new 10R15C tire to the same wheel.

3.4.4.5 CUCV-II. The mounter-demounter shall successfully break the bead and dismount a used LT235/85R-16E on/off road tire from a General Motors Wheel Number 14035374. For information purposes, the wheel has an eight hole lug pattern with 6.5 inch diameter bolt circle and a .67 inch offset. After breaking the bead and demounting the tire the mounter-demounter shall mount a new LT235/85R-16E on the same wheel. The CUCV is the Army's commercial utility cargo vehicle which is General Motors Corp., Chevrolet Blazer pick-up truck. Both the 15 inch and 16 inch wheels are available from commercial sources.

3.5 Detail of components. All components and component systems shall be designed for safe and efficient tire changing operations. The manufacturer may elect to use any system or combination of systems referenced within the specification to accomplish the required performance results.

3.5.1 Base The base shall be constructed of steel capable of supporting the mounter-demounter during the operations specified herein. The base shall incorporate a method for securing the mounter-demounter to the floor to retain the mounter-demounter in a stationary position during the tire mounting and demounting operation.

3.5.2 Motor and drive mechanism enclosure. The mounter-demounter shall incorporate a housing to enclose the drive mechanism. The housing shall be removable or shall include removable panels to afford access to the motor and drive.

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:19Z
Check the source to verify that this is the currentversion before use.

3.5.3 Wheel holding unit. The wheel holding unit shall mount and retain the wheel and tire assemblies, within the capabilities specified. Rotary type demounters shall be capable of demounting the tire in only one direction of rotation. Wheel holding units shall grip and retain the wheel by positive action such that the wheel shall not be loosened without deliberate positive action by operator. The wheel shall not be deformed or released from the grip of the wheel holding unit if, or when, the rotation is reversed with the tool jammed between the tire and wheel such that the tool prevents the wheel and tire from rotating in the reverse direction. Wheel and tire assemblies may be held on mounting spools with adapters for the different sizes of wheels to be accommodated.

3.5.4 Wheel lifting unit. The mounter-demounter shall incorporate a power lifting unit capable of lifting the wheel and tire assemblies specified in 3.4.4 thru 3.4.9 from floor lever, or by means of using a ramp, from a level elevated slightly above floor level, to the demounting position and lowering the wheel and rim assembly to floor or ramp level, after removal of the tire, in such a manner that the wheel is not driven into the floor or ramp. The lifting unit shall be operated by a foot or hand control.

3.5.5 Bead breaker unit(s). The mounter-demounter shall incorporate bead breaker unit(s) for breaking the beads away from the wheel. The bead breaker unit(s) shall break the outer and inner tire beads away from the wheel rim and shall completely remove tires from drop center rims. This shall be accomplished without the necessity of repositioning the wheel assembly on the wheel holding and rotating device. The bead breaker unit(s) shall not interfere with the mounting of the wheel and tire assembly on the wheel holding unit and shall incorporate a controlled positive adjustment to securely position the bead breaker unit(s) to accommodate the tire capacities specified herein.

3.5.6 Pneumatic system(s). Mechanisms utilizing compressed air as a power source shall function and perform with no more than 100 psig air pressure. Mechanisms that function and perform adequately to meet the requirements herein on less than 100 psig are also acceptable. Pneumatic pressure shall be maintained for not less than two minutes at a force of not less than 90 percent of operating pressure with no more than 5 psig pressure drop when the unit is pressurized and then isolated from the source of air pressure with a temporary valve placed in the air line for test purposes at the inlet coupling on the tire changer. Air hoses shall be of a proper length and design such that no kinks in the hoses shall occur during normal operation.

3.5.7 Hydraulic systems. Hydraulic system components shall be rated for three times the maximum operating pressure in the mounter-demounter. Hydraulic systems shall operate at maximum pressure for ten minutes in a dead stall condition without damage to the pump, all other components, and with no visible leakage.

3.5.7.1 Hydraulic gages. A 2 inch or larger hydraulic gage shall be placed in easy view of the operator that accurately indicates system pressure within one percent of full scale value. The full scale value of the gage shall not exceed the maximum system operating pressure by more than one hundred percent.

3.5.8 Electrical systems. The electrical components shall conform to applicable NEMA Standards, unless otherwise specified herein. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured to prevent loss of tightness. Electrical soldering shall conform to DOD STD 2000-4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:19Z
Check the source to verify that this is the current version before use.

3.5.8.1 Motor. Motors for driving the mounter-demounter shall be of sufficient horsepower to meet the requirements specified herein. Unless otherwise specified (see 6.2), the drive motor shall be an induction squirrel-cage type of a continuous-duty classification, with an ambient temperature rating of 50 degrees centigrade in conformance with NEMA MG-1, with Grade B insulation or better. Unless otherwise specified (see 6.2), the motor shall be 208/230 volts, 50/60 Hertz (Hz), single-phase. The motor shall be protected against thermal damage from electrical overload with a manual reset protection device.

3.5.8.2 Electric power supply cable. An electric power supply cable shall be installed on each machine. Unless otherwise specified, the cable shall be not less than 10 feet in length and shall terminate in not less than one inch of stripped tinned wire without a receptacle plug. The cable covering shall be type SO heavy duty, in accordance with ANSI C1.

3.5.8.3 Electrical protection. The mounter-demounter's electrical circuits shall be protected against thermal overload above 50 degrees centigrade or short. The thermal overload protection device shall be manually resetable.

3.5.8.4 Power switch. An electric power on-off switch shall be provided and may include the thermal overload device.

3.5.9 Controls. Hand or foot operated controls that control the movements of the machine shall be of the "deadman" type, returning to neutral position when released, thereby stopping the movement of the machine.

3.6 Dielectric strength. The conductors and all electrical components shall withstand, without damage, twice the rated voltage plus 3000 volts root mean square (RMS) at 60 Hz for a period of not less than one minute between mutually insulated circuits, and between ground and insulated circuits.

3.7 Hand tools. The mounter-demounter shall be delivered with all tire irons necessary to perform tire changing including split ring removal.

3.8 Identification marking. Unless otherwise specified (see 6.2), a corrosion-resistant metal name plate in accordance with MIL-P-19834 shall be permanently and legibly marked in accordance with MIL-STD-130, including National Stock Number, and shall be securely attached to each mounter-demounter.

3.9 Workmanship. Standards of workmanship shall assure that the mounter-demounter shall have the strength, safety, stability, and efficient operating characteristics found in standard commercial units and as specified herein.

3.10 Repair part manuals. Repair parts and accessory identification in repair part manuals shall include the prime contractor's name, nomenclature, part number. Repair part identification in repair part manuals shall also include the original equipment/part manufacturer's (OEM) name, Commerical And Government Entity (CAGE), nomenclature and part number of the part used in the end item, immediately after the prime contractor's part identification. Items with high mortality rates shall be so identified. See paragraph 3.1.2.2 thru 3.1.2.10 (especially paragraph 3.1.2.8.2.2) of MIL-M-7298C. Repair parts listings shall not indicate or state that any equipment required to comply with contract performance and compliance is "optional", "accessory", or should be ordered separately.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:19Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products submitted to the Government for acceptance comply with all requirements of the contract. Sampling for quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When specified, first article inspection shall be performed on the preproduction model(s) or initial production item(s) noted in 3.1. First article inspection shall consist of the examination in 4.5, and the tests of 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production units offered for acceptance under the contract. Quality conformance inspection shall consist of a through f as follows, and failure of any unit to pass an examination or test shall be cause for rejection of the unit.

a. Product examination (see 4.5)
b. Operational test (see 4.6.1)
c. Pressure unit leak test (see 4.6.2)
d. Jammed tool test (see 4.6.4)
e. Dielectric strength test (see 4.6.5)
f. Packaging inspection (see 4.7)

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed in accordance with MIl-STD-105, using Inspection Levels and Acceptable Quality Levels (AQL) as listed, unless 100 percent inspection is specified. A sample lot shall consist of a quantity produced in one month or 80 mounter-demounters, whichever is less.

Inspection or test	Level	AQL
Product examination	II	1.0
Operational test	II	1.0
Pressure unit leak test	S-4	1.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:19Z
Check the source to verify that this is the current version before use.

Jammed tool test	S-4	2.5
Dielectric test	S-4	4.0
Packaging inspection	S-4	4.0

4.5 Product examination. Visually, dimensionally, and manually examine each mounter-demounter to determine conformance with the requirements of 3.2 through 3.3.10, 3.5 through 3.5.9 and 3.7 through 3.10. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from identified defects. Dimensional examination includes measuring dimensions and weights, as specified. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the mounter-demounter and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection (see 3.3.8). In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item, shall be cause for rejection.

4.6 Tests. Failure of the mounter-demounter to pass any of the listed tests shall be cause for rejection.

4.6.1 Operational test. The fully assembled mounter-demounter shall be set up for operation in accordance with the manufacturer's instructions and turned on. All mounter-demounter controls shall be fully exercised to put the machine through its full range of motions, and the operation of the mounter-demounter observed for proper response.

4.6.2 Pressure unit leak test. For those units utilizing a pressure unit, the pressure unit(s), either before or after assembly to the mounter-demounter, shall be connected in series with an air pressure gage and a leak-free shut-off valve to a regulated source of compressed air. The air pressure gage shall be located between the shut-off valve and the pressure unit(s), and the assembly shall be as direct and compact as possible to minimize the volume of air which can be entrapped between the shut-off valve and the pressure unit(s). The pressure gage shall be accurate within one percent of the reading over the range of 75 to 125 psig and readable to the nearest psig. The gage shall be calibrated, and the calibration standard shall be traceable to the National Bureau of Standards. The source of compressed air shall be capable of providing regulated pressures of from 0 to 100 psig or more. With the shut-off valve open, the supply pressure shall be raised to 90 psig or more and the shut-off valve closed to entrap the compressed air in the pressure unit(s). The pressure gage shall then be observed for a period of not less than two minutes. Air leakage from any pressure unit sufficient to cause a drop in pressure of more than five psig over the period of two minutes shall be cause for rejection (see 3.5.6).

4.6.3 Performance test. The mounter-demounter shall demount and mount all tires as listed in paragraphs 3.4.4.1 through 3.4.4.6. Each tire and wheel assembly shall be positioned on the machine using the wheel lifting unit (see 3.5.4) and held in place with the wheel holding unit (see 3.5.3). With the tire deflated, the bond between each bead and rim shall be broken by the bead breaker unit(s) (see 3.5.5), and the tire removed from the wheel. The removed tire or a new tire, as specified, shall then be mounted, forcing both beads over one side of the rim. The wheel and tire assembly shall be released, lowered to the floor, and removed from the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:19Z
Check the source to verify that this is the current version before use.

mounter-demounter. The mounter-demounter, tires, and wheels shall be examined for damage during and after each demounting and mounting. After mounting, each tire shall be inflated to its rated pressure and tested for leakage. Leakage past either bead shall be considered evidence of damage. Failure of the mounter-demounter to demount and mount all tires listed in 3.4.4.1 through 3.4.4.6 without damage to itself, and without damage to the tires and wheels, shall be cause for rejection.

4.6.4 Jammed tool test. The fully assembled mounter-demounter shall be set up such that a wheel and tire is mounted with the shoe inserted and partially breaking the bead. Under this condition, locking pliers, C-clamp or other device shall be attached to the wheel so that they block the shoe, preventing rotation of the wheel and tire in the reverse direction. The wheel holding unit will be reversed until contact is made with the locking pliers. Upon binding for 30 seconds, the wheel holding unit shall not loosen the wheel. Loosening of the wheel, or deformation of the wheel under this condition (see 3.5.3), shall be cause for rejection. Failure of the mounter-demounter to demount and mount each tire within the 20 minute time frame as specified in 3.4.1 shall be cause for rejection.

4.6.5 Dielectric strength test. The electrical circuits shall be subjected to 3460±40 volts rms at 60 Hz applied between mutually insulated circuits, between insulated circuits and ground, and between each conductor and the chassis ground (see 3.6). Voltage shall be increased from zero to test voltage at the approximate rate of 500 volts per second, held at test voltage for one minute and removed at the same rate as applied. During the one minute at test voltage the circuitry shall be observed for flashover (surface discharge), sparkover (air discharge) and breakdown (puncture discharge). Any evidence of flashover, sparkover, breakdown, or damage to the unit due to heat or other effects of the applied voltage, shall be cause for rejection.

4.7 Packaging inspection. The mounter-demounter shall be inspected before and after packaging to determine compliance with the preservation, packaging, packing, and marking requirements specified in Section 5 of this specification.

5. Packaging.

5.1 Disassembly. The mounter-demounter may be disassembled to the extent necessary to assure that all surfaces requiring a preservative are accessible for processing, to assure minimum cube for packing, and to permit reassembly without special skills or tools. Disassembly, when applicable, shall be as specified in MIL-STD-1186.

5.2 Preservation. Preservation shall be Level A or Level C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. The mounter-demounter shall be cleaned and dried in accordance with any applicable process and procedure of MIL-P-116.

5.2.1.2 Preservative application. Immediately after cleaning and drying, apply a coat of type P-2 or type P-11 preservative to all exterior, critical, ferrous metal surfaces of the mounter-demounter and accessories. Exterior noncritical, ferrous metal surfaces which have not been painted or plated shall be coated with type P-1 or type P-19 preservative. Interior surfaces of the air motor shall be coated with a preservative oil conforming to type P-10, grade 30. The oil shall be applied by injection into the air stream while operating the motor until the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:19Z
Check the source to verify that this is the current version before use.

preservative appears at the exhaust port. If the hydraulic preservative oil is not compatible whit the agency for the selection of an alternate preservative. P-type preservatives specified shall conform to MIL-P-116.

5.2.1.3 Unit packs. The method of unit protection shall conform to method I of MIL-P-116. Preserved critical, ferrous metal surfaces shall be wrapped with barrier material conforming to MIL-B-121, type II, grade A, class optional. The hydraulic ram shall be secured in a retracted position. Air inlets and outlets shall be sealed with plastic caps or plugs conforming to MIL-C-5501 or tape conforming to PPP-T-60, class IV. The electric motor shall be packaged in accordance with level A requirements of MIL-E-16298 and the applicable subparagraph for assembled machines. The pressure gage, as applicable, shall be covered with one inch thick uncompressed cushioning material conforming to PPP-C-843, type II. Secure cushioning and barrier material in place with tape conforming to PPP-T-60, type optional. Technical data issued with the unit shall be packaged in accordance with method IC-1 of MIL-P-116. Coil the power supply cord to a minimum safe diameter and secure to the basic unit with cotton tape or twine. The connector plug shall be protected in accordance with method III of MIL-P-116.

5.2.1.4 Consolidated package. Consolidate the packaged tools, technical data, and all small disassembled components and accessories. Package in a snug-fitting fiberboard box conforming to PPP-B-636, class weather-resistant. Cushioning material specified in 5.2.1.3 shall be employed to restrict movement within the box

5.2.2 Level C. Level C preservation shall be as specified in MIL-STD-1190.

5.3 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.3.1 Level A. Each mounter-demounter shall be packed in a box conforming to PPP-B-621, class 2, or a crate conforming to MIL-C-52950, type V, style B, covered. the selection of the container shall be governed by the limitations of the container specifications. The consolidated package (see 5.2.1.4) shall be placed inside the shipping container and secured to restrict movement. The contents of an open crate shall be covered with a waterproof barrier shroud. Anchoring, shrouding, blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186.

5.3.2 Level C. Level C packing shall be as specified in MIL-STD-1190.

5.4 Marking. All marking shall be as specified in MIL-STD-129. Bar Code marking is also required and shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The mounter-demounter is intended for use in base shops, field service, and maintenance shops to facilitate the changing of tires on passenger cars, military vehicles, and light and heavy commercial trucks and trailers.

6.2 Acquisition requirements. Acquisition documents should specify the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:19Z
Check the source to verify that this is the current version before use.

a. Title, number, and date of this specification.
b. First article when required (see 3.1).
c. Special safety devices when required (see 3.3.7).
d. Painting if different (see 3.3.10).
e. Motors if different and current, voltage, Hz, and phase if different (see 3.5.8.1).
f. Power supply cord if different (see 3.5.8.2).
g. Identification marking if different (see 3.8).
h. Level of preservation, packaging, packing and marking required (see 5.2, 5.3, and 5.4).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operations and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements should be specified by the user.

6.5 Subject term (key word) listing.

AMSC number
Design
First article
First article inspection
Materials
Ordering data
Performance
Protective finish
Scope

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - YD
Air Force - 99

Preparing activity:
Army - AL

Project No. 4910-0675

Review activities:
Army - AT
Navy - None
Air Force - None

Source: https://assist.dla.mil - Downloaded: 2026-10-22T02:19Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:19Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-M-45083F, Mounter-Demounter, Pneumatic Tire, Stationary

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:19Z
Check the source to verify that this is the current version before use.

necessary retaining power on the end

six circular corrugations to provide the

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Armament Research Development
and Engineering Center
ATTN: SMCAR-EST-S
Rock Island, IL 61299-7300

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:19Z
Check the source to verify that this is the current version before use.